Sumitomo Wiring Systems
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WIRELESS JIG DETECTION

Eliminating Wrong Dimensions, Ensuring Traceability, and Improving Quality

SEWS-Eg
Presented by:
Future Industry Team (Team 14)
Date: April 13, 2026

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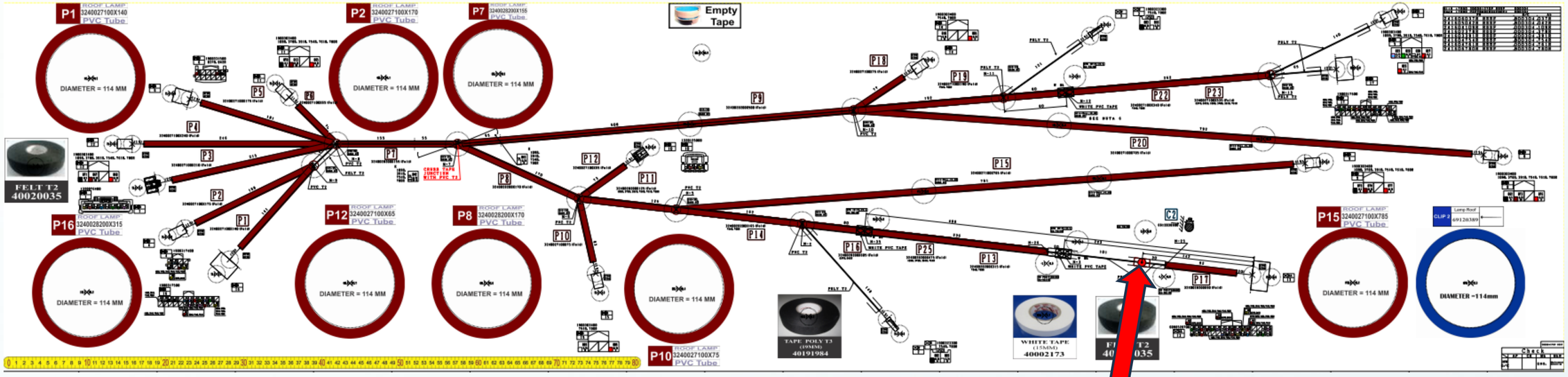
DMAIC: DEFINE

- Assembly boards on carousels with no clip boards generally have no jig detection, and so no traceability of whether the connectors and clips were correctly used as per SEWS standards.



Clip Without Detection

Confidential

FUTURE INDUSTRY TEAM

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Quality Supervisor

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Maintenance Operator

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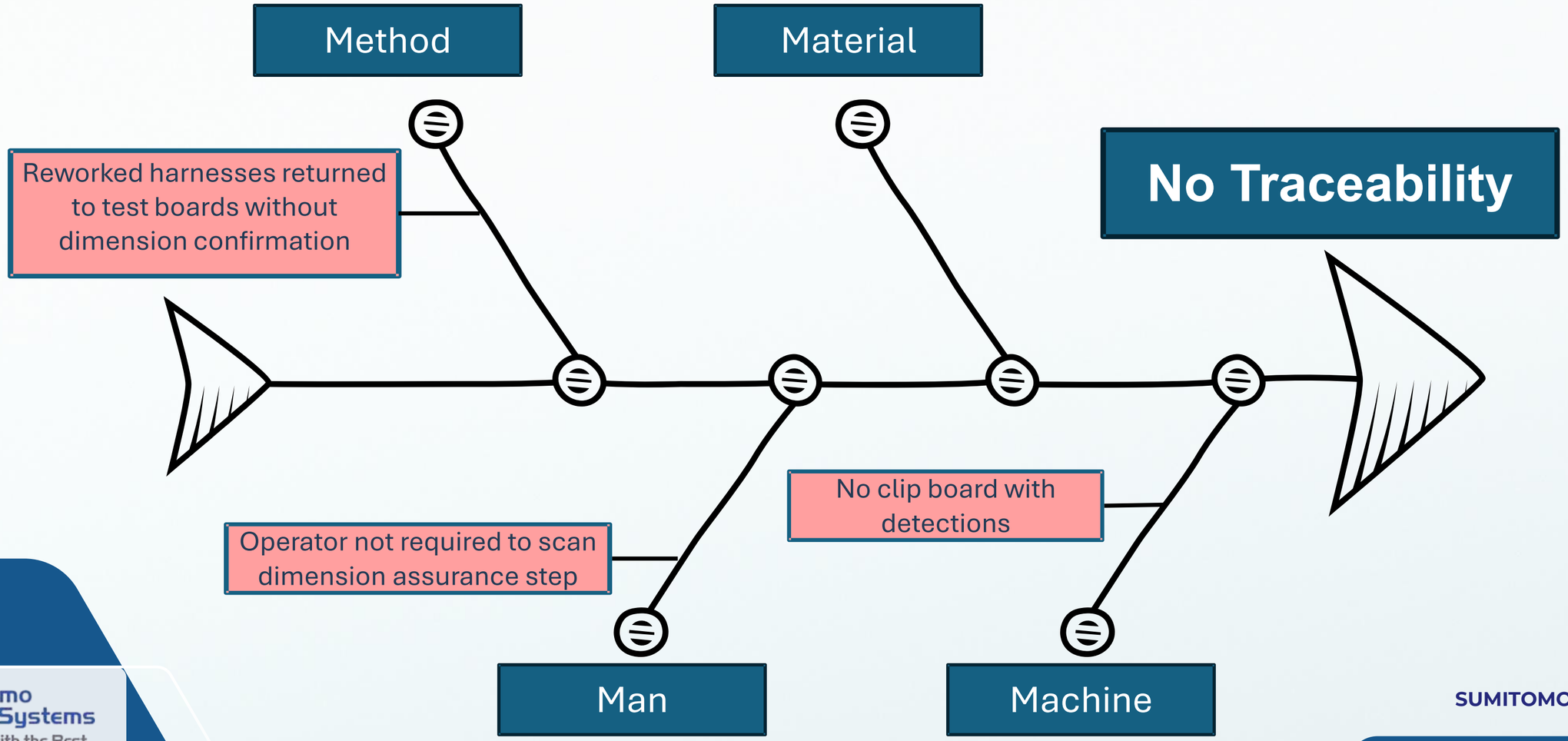
DMAIC: MEASURE

- There are several metrics we can use to quantify and map the process to determine if the proposed solution is effective such as the number of compliance reports related to misuse of jigs, wrong dimension defects, and rework rates

- In the case of B1318 Roof, an average of 156 reworked harnesses per month have been recorded, and each harness returns from rework to electric test board for presence without dimension confirmation.

DMAIC: ANALYZE

- **Fishbone Diagram**



DMAIC: IMPROVE

- **Methodology**

Board ID#10 ~----

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Board ID#05 ~----

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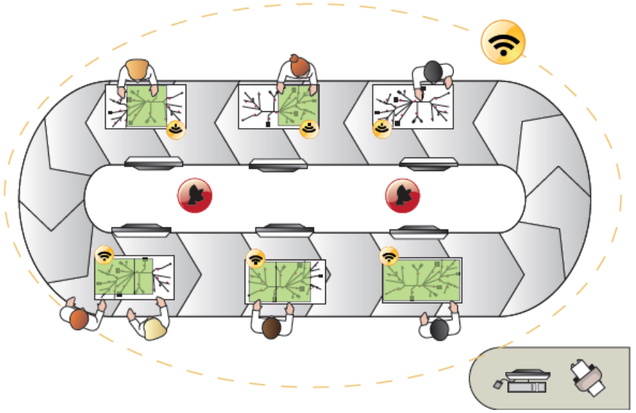
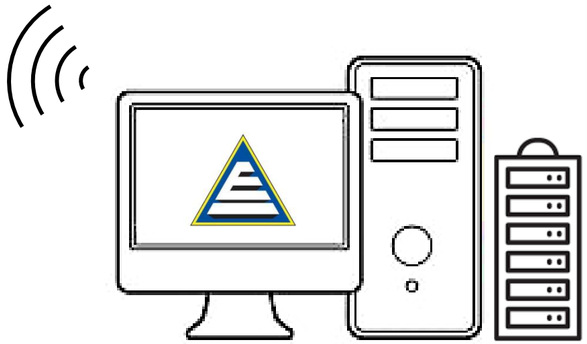
Board ID#01 ~----



Communication link via RF-waves

Data transmission and traceability

Link between manufacturing and Quality



Transmission side

Receiver Side

DMAIC: IMPROVE

- Test Sequence

VIDEO

DMAIC: IMPROVE

- **Planned/Future Improvements**
  - Per board Activation Switch(implemented)
  - Board Lock Process implementation(implemented)
  - System Traceability Implementation
  - S/S maintenance check sheet
  - Compact Hardware design (SERS Raised)

DMAIC: IMPROVE

- ## Cost Analysis

The system is designed connect existing test equipment and software on SEWS-Eg plants without the need of extra dedicated software and hardware to combine carousel and clip board.

Items	European supplier	Improvement project
Test jigs	~87€ per jig	~10€ per jig
Test Hardware (Tx/Rx)	725€	300€
Test Software	New (280 Euro)	Compatible with current system
Spare parts	52€	7€

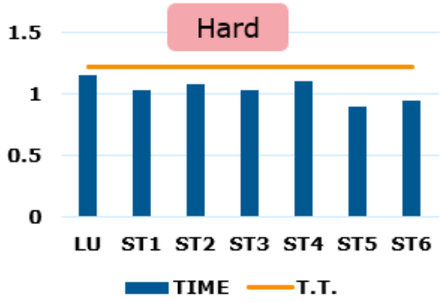
DMAIC: IMPROVE

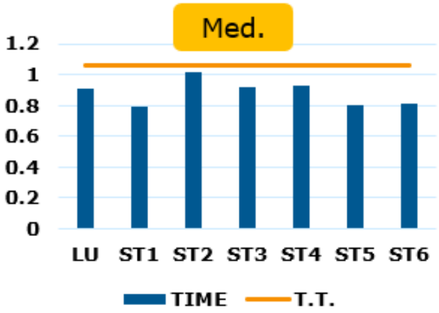
1- Time Study To The Carousel Stations:

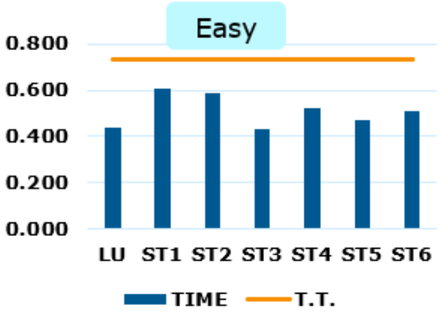
Process	TIME	T.T.	Free Time
LU	1.15	1.223	0.073
ST1	1.019	1.223	0.204
ST2	1.068	1.223	0.155
ST3	1.030	1.223	0.193
ST4	1.103	1.223	0.1205
ST5	0.890	1.223	0.333
ST6	0.940	1.223	0.283

Process	TIME	T.T.	Free Time
LU	0.909	1.059	0.15
ST1	0.798	1.059	0.261
ST2	1.017	1.059	0.0425
ST3	0.922	1.059	0.1375
ST4	0.931	1.059	0.128
ST5	0.800	1.059	0.259
ST6	0.810	1.059	0.249

Process	TIME	T.T.	Free Time
LU	0.437	0.735	0.298
ST1	0.608	0.735	0.127
ST2	0.590	0.735	0.145
ST3	0.430	0.735	0.305
ST4	0.520	0.735	0.215
ST5	0.470	0.735	0.265
ST6	0.510	0.735	0.225







2- Estimation time for required data:

1- Estimation time for scanning travel label.
2- Estimation time for scanning board label.
3- Estimation time for printing label after finishing the program and attach it to the harness.

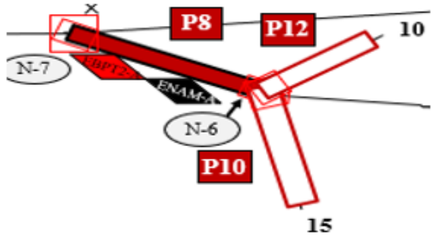
scan travel label	0.037
scan board label	0.037
printing label & attach it to harness	0.24
Total Required Time	0.314

DMAIC: IMPROVE

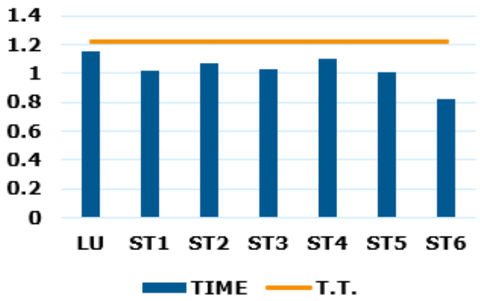
3- Free Up Time To Station 6 (Time After):

Hard

Moving the two locals of fixing sheets for branch.10 & branch 15 to be on station 5.

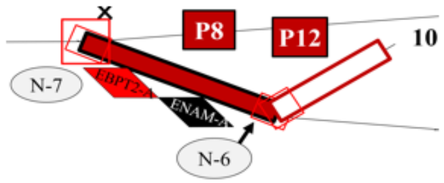


Process	TIME	T.T.	Free Time
LU	1.15	1.223	0.073
ST1	1.019	1.223	0.204
ST2	1.068	1.223	0.155
ST3	1.03	1.223	0.193
ST4	1.1025	1.223	0.1205
ST5	1.012	1.223	0.211
ST6	0.818	1.223	0.405

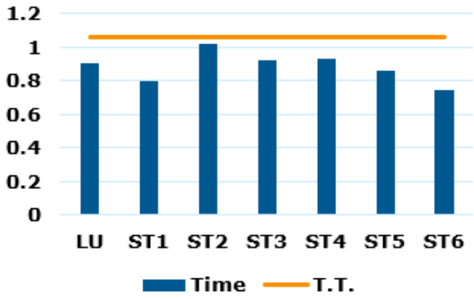


Med.

Moving the local of fixing sheets for branch.10 to be on station 5.

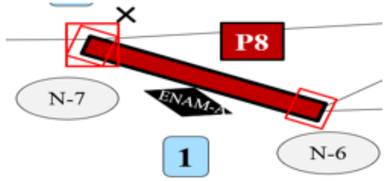


Process	TIME	T.T.	Free Time
LU	0.909	1.059	0.15
ST1	0.798	1.059	0.261
ST2	1.017	1.059	0.0425
ST3	0.922	1.059	0.1375
ST4	0.931	1.059	0.128
ST5	0.861	1.059	0.198
ST6	0.749	1.059	0.31

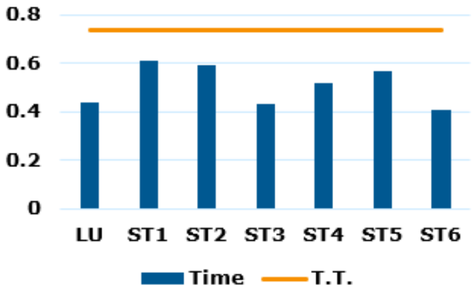


Easy

Moving the local of assembly and putting sheet without fixation.



Process	TIME	T.T.	Free Time
LU	0.437	0.735	0.298
ST1	0.608	0.735	0.127
ST2	0.59	0.735	0.145
ST3	0.43	0.735	0.305
ST4	0.52	0.735	0.215
ST5	0.57	0.735	0.165
ST6	0.41	0.735	0.325



DMAIC: CONTROL

• Risk Assessment

Risks associated with failure of such systems has been identified and addressed:

1. **Radio Frequency Interference**
   - Tested with multiple switches next to each other.
   - Tested with walkie-talkie radio interference nearby.

2. **Battery Failure**
   - Lithium-ion batteries cheap and readily available.
   - Expected battery life can be calculated and prepared.

3. **Hardware Damage**
   - Switches used are the same as the existing switches (cheap and available)
   - Transmitters, receivers, and encoders are locally supplied and cheap.

DMAIC: CONTROL

- **Control Plan**

After full implementation on one line, the following KPIs will be monitored and compared to evaluate the project impact:

- Rework rates
- Wrong dimension defects
- Number of compliance reports

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Thank You
For Your Attention

Presented By
Future Industry Team (Team 14)

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SWS WAY

Action principles for all of us in the Sumitomo harness business

PROFESSIONALISM

•Customer satisfaction •Honesty & Integrity •Pride in what we do

TEAMWORK

•Mutual communication •Accountability & Cooperation •Diversity & Inclusion

CHALLENGE

•Forward thinking •Learning from experience •Ambition & Perseverance